

Activity – Example 2

The frame assignment is not unique:

- There are two choices of direction for $\hat{Z}_i$.
- For intersecting axes, the sense of direction of the mutual perpendicular is arbitrary and thus there are two choices of direction for $\hat{X}_i$.
- For parallel axes, the origin of frame $\{i\}$ is arbitrary.
- For prismatic joint, there is also some freedom in assigning the frame.

In the video (and lecture notes), we have assigned the origin of frame $\{3\}$ at the same location as the origin of frame $\{2\}$ (Figure 1), from which we derived 0P . We then stated that the end effector is at position of ${}^3P = [0 \ 0 \ L_2 + L_3]^T$, and we calculated ${}^0P = {}^0_3T \cdot {}^3P$.

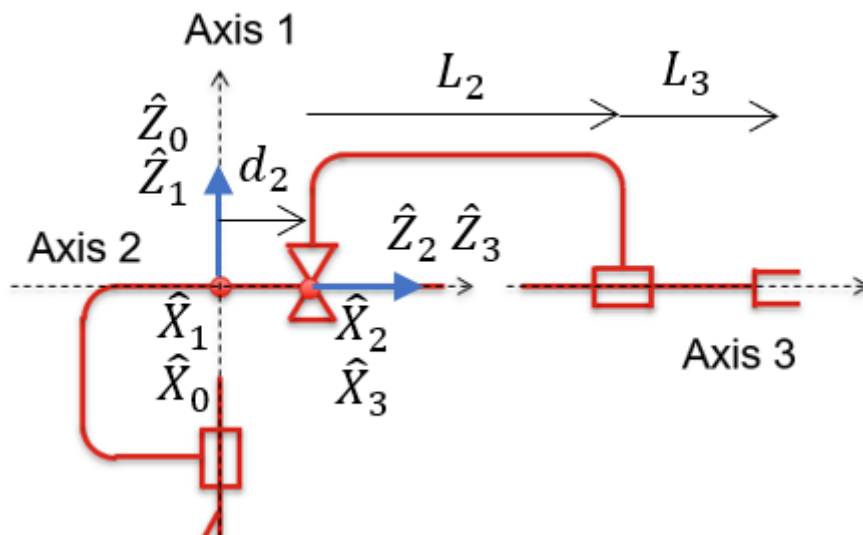


Figure 1: Frame $\{3\}$ at same position as frame $\{2\}$

However, the definition of X_3 only specifies that the direction of X_3 is aligned with X_2 when $\theta_3 = 0$. If we put frame $\{3\}$ at the third joint as shown in Figure 2, the above definition is still valid, because if we view the robot from the side (Figure 3), X_3 is indeed aligned with X_2 .

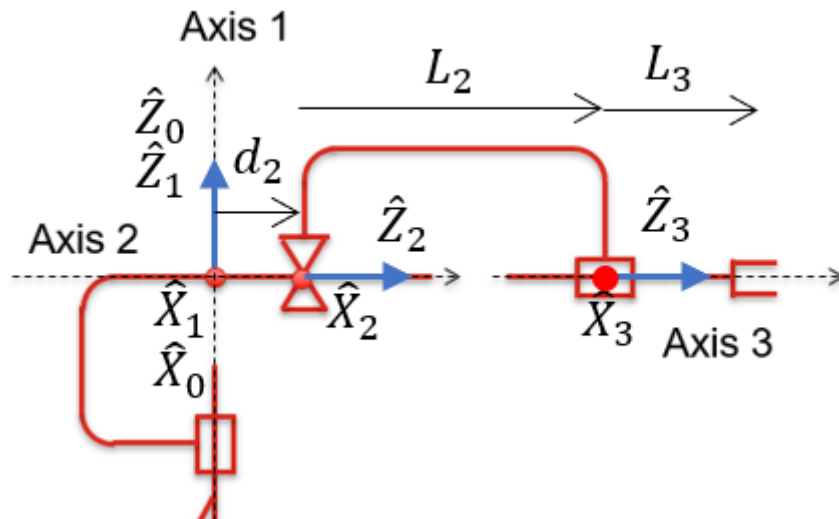


Figure 2: Frame {3} at location of physical joint

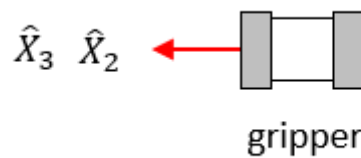


Figure 3: X_2 aligned with X_3 from the side view

Exercise:

With the new frame assignment as shown in Figure 2, determine the DH parameters, the *new* position of end-effector with respect to frame {3}, and the *new* position of end-effector with respect to frame {0}.

Is the new 0P the same as the one derived in the lecture video / slides?

Expected answer:

Yes, the answers should be the same. With the new placement of {3}, d_3 as well as 3P will be different from what was shown in lecture video / slides, but the result of ${}^0P = {}^0T \cdot {}^3P$ will be the same.